Alireza Jaberi Rad

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SUMMARY

PhD candidate in Computer Engineering specializing in hardware–software co-design and brain-inspired accelerators for edge intelligence. Hands-on experience in algorithm development, system-level design, and FPGA programming, with publications at NeurIPS and Nature-family journals. Demonstrated track record leading student teams, collaborating on cross-functional research, and shipping projects end-to-end.

EDUCATION

University of Massachusetts Amherst Sep 2023 – May 2028 (Expected)
Ph.D. in Computer Engineering Amherst, MA

University of Tehran Sep 2017 – Jun 2022
B.Sc. in Computer Science | B.Sc. in Electrical Engineering (dual degree) Tehran, Iran

TECHNICAL SKILLS

Languages: Python, C, C++, Verilog, MATLAB, SystemC

ML / AI: PyTorch, NumPy, Pandas, Matplotlib, neural networks, hyperdimensional computing

FPGA & Embedded: Intel/Altera FPGAs, Xilinx FPGAs, Quartus, Qsys, ST & Arduino microcontrollers

IC & Hardware Design: Cadence Virtuoso, Cadence Innovus, HSPICE, LTspice, Altium Designer

Tools & Workflow: Git, Linux, VS Code Live Share, LaTeX, Tetramem SDK

Soft Skills: Technical leadership, cross-functional collaboration, project management

RESEARCH EXPERIENCE

Analog Twin for Digital Circuits Dec 2024 – Present
Biologically Inspired Neural and Dynamical Systems Lab, UMass Amherst Amherst, MA

- Converted digital logic into neural-network surrogates mapped onto an analog memristive accelerator for low-power inference.
- Designed an analog activation function and verified end-to-end behavior in SPICE simulation.
- Built a Python simulator of memristor non-idealities (cycle-to-cycle and device-to-device variation) for accurate analog-inference predictions.
- Built a 28 nm CMOS framework benchmarking analog twins against digital counterparts on speed, area, and energy.
- Engineered a verification flow for analog-twin functionality on a Tetramem memristive system-on-chip (SoC).

Hyperdimensional Computing – Hardware/Algorithm Co-Design May 2024 – May 2025
Published at NeurIPS 2024 | Journal extension under review at Nature Communications

- Designed an HDC algorithm that turns memristive in-memory stochasticity into a computational feature.
- Implemented and benchmarked the algorithm on a Tetramem memristive SoC in Python and PyTorch via the Tetramem SDK.
- Achieved state-of-the-art accuracy on a language-classification benchmark: **99.35%** (software) / **95.24%** (hardware).
- Reduced hardware resource usage by **90%** via tight algorithm–SoC co-design.

3D Memristive Arrays for Single-Step Matrix Operations Sep 2023 – Jan 2026
Journal paper under review at Advanced Intelligent Systems

- Architected a stackable 3D memristive crossbar array, inspired by the human retina, that ingests 2D images directly without serialization.
- Demonstrated one-shot ($O(1)$) matrix multiplication and matrix–kernel convolution in a single analog read.
- Verified the design on a CMOS prototype, reaching **92.96%** precision on an edge-detection benchmark.
- Fabricated a nanoscale memristive 3D array, achieving **89.31%** edge-detection precision in silicon.
- Delivered **365**$\times$ lower energy and orders-of-magnitude lower latency vs. an equivalent CMOS digital implementation.

WORK & TEACHING EXPERIENCE

Summer Engineering Institute (SENGI) Supervisor Jul 2024
University of Massachusetts Amherst Amherst, MA

- Led a cohort of K–12 students in building a UI for a hyperdimensional-computing software simulation.
- Taught fundamentals of neural networks and hyperdimensional computing to a non-specialist audience.
- Coached students on clean code and Python tooling: NumPy, Pandas, Tkinter, Matplotlib.
- Mentored team-based development workflows in VS Code Live Share, simulating an industry collaboration setting.

FPGA-Based Audio Processing Accelerator Jan 2021 – Jun 2021
Embedded Systems Laboratory, University of Tehran Tehran, Iran

- Built a hardware accelerator for real-time audio signal processing and denoising on an FPGA board.
- Implemented the system on an Intel/Altera Cyclone II FPGA in Verilog and C++ using Quartus and Qsys.
- Achieved **100**× speedup over a fully software baseline.
- Shipped a UI to record/play/echo audio and visualize sound intensity across multiple time intervals.

SELECTED PUBLICATIONS

A. Jaberi Rad, et al. "Single-Step Analogue In-Memory Matrix Computation in Three-Dimensional Circuits." *Advanced Intelligent Systems* (under review).

Y. Huang, **A. Jaberi Rad**, et al. "Hardware–Algorithm Co-Design for Hyperdimensional Computing Based on Memristive System-on-Chip." *NeurIPS 2024*.

Y. Huang, **A. Jaberi Rad**, et al. "Hyperdimensional In-Memory Computing with Analogue Memristive Crossbar Arrays." *Nature Communications* (under review).

W. Zhao, Y. Huang, A. Tewari, **A. Jaberi Rad**, et al. "Event-based Neuromorphic Sensing System with Flexible Haptic Sensors and a Memristive System on a Chip." *Nature Sensors*.

C. He, Y. Huang, A. Korkmaz, **A. Jaberi Rad**, et al. "Cognitive Radio Receiver with Analogue In-Memory Computing." *Science Advances* (under review).

Y. Ling, Z. Xiong, Y. Huang, **A. Jaberi Rad**, et al. "Tunable Diffusive Memristor Arrays for Neuromorphic Temporal Computing." *Nature Materials* (under review).

HONORS & AWARDS

David H. Navon fellowship award for outstanding research contribution in the field of microelectronics.	2025
Second-place winner in the ECE Graduate Research Poster Session, UMass Amherst.	2025
Best Undergraduate Project Award, University of Tehran.	2022
Exceptional-talent M.Sc. admission award for Sharif University of Technology and University of Tehran.	2022
Top 10% of both Computer Science and Electrical Engineering cohort, University of Tehran.	2019 – 2022
FoE Award; ranked 3$^{\text{rd}}$ / 40 in Computer Science and 11$^{\text{th}}$ / 135 in Electrical Engineering, University of Tehran.	2019
Awarded a tuition-free second major (only 103 of 32,000 University of Tehran students qualified).	2018
Ranked 37$^{\text{th}}$ of 150,000 in the Iranian National University Entrance Examination.	2017
Semi-finalist in both Iran's National Mathematics Olympiad and National Computer Olympiad.	2015 – 2016
Three-time karate national champion of Iran; undefeated in Khuzestan province.	2005 – 2014

RELEVANT COURSEWORK

Electronic System-Level Design, VLSI, FPGA Design, Computer Architecture, Neuromorphic Engineering, Data Mining, Artificial Intelligence, Advanced Programming, Engineering Leadership & Entrepreneurship, Project Management.